

# Predicting Value with Accounting Data

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Source: <https://github.com/tejasmanojav/P-B-and-Friends>

## Introduction

I want to see if raw accounting data can reliably show if a company is under or overvalued. This helps cut through daily market noise.

I do not want to guess market sentiment or investor psychology. Instead, I use simple code. It checks if financial statement signals can systematically find strong but underpriced stocks.

My primary tool is the Price-to-Book (**P/B**) ratio. This ratio connects a company's stock price directly to its **Book Value of Equity**. The book value represents the net asset value on its balance sheet:

$$\text{Book Value of Equity} = \text{Total Assets} - \text{Total Liabilities}$$

Financial accounting relies on historical cost. Corporate assets are typically recorded at their original cost minus accumulated depreciation. This avoids speculative market estimates. Book Value acts as a grounded, real-world floor for what a company is worth on paper. It stays independent of stock market hype.

I divide a company's total market capitalization by this concrete baseline to find the **P/B ratio**. A stock trading at a low **P/B ratio** near or below 1.0 is a "value stock." This idea is compelling. A low ratio suggests the stock is very cheap relative to its actual net assets. This offers a built-in "margin of safety" and protects investors from severe downside risk.

But this traditional model has a major catch. A low **P/B ratio** can signal a **value trap**. A company might look cheap only because its business is failing or buried in debt. To avoid this, my pipeline adds profitability metrics to the initial **P/B** filter. I use **Return on Equity (ROE)** and **Return on Assets (ROA)**. I evaluate how effectively a management team generates net income from equity and assets. This helps filter out deteriorating businesses and find actual value opportunities.

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## Research Objective and Hypothesis

I want to see if simple accounting data can beat market noise. To test this, I built a strategy based entirely on company fundamentals. First, I use the P/B ratio to find cheap stocks. Then, I add ROA and ROE filters to make sure these companies are healthy and profitable. This helps me avoid dangerous value traps.

My key idea is that combining value and quality filters will reveal stocks that the market misprices. If I am right, my chosen basket of stocks should regularly beat the standard sector index.

## Empirical Evaluation

Because financial time-series data is notoriously unpredictable and violates the clean assumptions needed for traditional p-value tests, this study evaluates the hypothesis using two practical, empirical metrics:

- **Absolute Performance ( $\mu$ ):** I track whether the strategy's basket achieves a higher average monthly return ( $\mu_{\text{Strategy}}$ ) than the reference sector index ( $\mu_{\text{Sector}}$ ) over the backtest period.
  - $\mu_{\text{Strategy}} > \mu_{\text{Sector}}$
  - **Risk-Adjusted Alpha ( $\alpha$ ):** I run a linear regression of my strategy's returns against the reference sector index. This allows me to calculate **Alpha ( $\alpha$ )**—the exact return premium generated by my accounting filters that cannot be explained by general sector movements. A successful strategy requires a positive alpha.
  - $\alpha > 0$
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## Data Sources

I split my data collection into two separate parts. One part is for accounting data. The other part is for market prices. Keeping them separate helps me set my accounting filters before I look at stock prices.

### Fundamental Accounting Data (SimFin)

I used the SimFin Python API to get past balance sheets and income statements. I pulled this data for all companies in my test. I took out net income, total assets, and total equity from these files. Then I used them to calculate my main metrics: P/B, ROE, and ROA. SimFin gives me clean data. This makes it easy to compare companies in different sectors.

### Market Pricing Data (yfinance)

I used the `yfinance` library to get stock prices from Yahoo Finance. I picked the "Adjusted Close" price. This price includes stock splits and dividends. This keeps artificial price drops

from messing up my test results. I needed this data to track how my portfolio performed, find market values, and build sector indexes.

## Investment Universe and Benchmarks

To test how well my accounting-informed factor strategy works across the entire economy, I evaluated it across 11 different stock market sectors. Instead of comparing my results to a broad index like the S&P 500, I matched each sector with its specific benchmark Exchange-Traded Fund (ETF) to ensure a direct, fair comparison.

The 11 sector-to-benchmark mappings evaluated are:

**Technology:** XLK; **Healthcare:** XLV; **Financial Services:** XLF; **Consumer Defensive:** XLP; **Consumer Cyclical:** XLY; **Industrials:** XLI; **Energy:** XLE; **Basic Materials:** XLB; **Real Estate:** XLRE; **Utilities:** XLU; **Business Services:** IYZ.

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## Methodology and Backtest Architecture

I built a Python colab to test my hypothesis. It turns accounting data into a trading strategy. The program has four parts. These are the selection algorithm, data matching, evaluation metrics, and the automation loop.

### Custom Index Construction Algorithm

The main algorithm filters stocks from a major sector ETF, like XLK for Tech. It does not pick random stocks. It uses the actual ETF holdings as a starting baseline. The selection follows four steps:

1. **Find the Stocks:** The script looks up the exact list of stocks in the ETF for any past date.
2. **Filter for Quality:** The script checks the financial data for each stock. It drops a company if its ROE or ROA is below the sector average. This avoids “value traps”.
3. **Rank by Value:** The script sorts the healthy stocks by their Price-to-Book (P/B) ratios. It ranks them from lowest to highest.
4. **Cap the Basket:** The script uses a strict cut-off size. It selects only the top K cheapest, high-quality stocks for the Custom Index.

### Data Alignment

Matching accounting data with stock prices was not straightforward. Companies do not release financial data on the exact day a quarter ends. For example, a quarter might end on December 31. However, the public sees the reports weeks or months later. I had to prevent look ahead bias (“looking into the future”).

To fix this, I added a strict three-month delay to all financial data. Suppose my script runs a test on April 1. It only looks at financial statements from December or earlier. This delay keeps the backtest realistic. It only uses data that was public at that moment.

## Empirical Performance Evaluation

I built an evaluation module to see if my filters beat the benchmark. It calculates five main indicators at the end of each run. These metrics compare my Custom Index against the baseline ETF:

- **CAGR (Compound Annual Growth Rate):** This measures the average annual return. It shows how fast the strategy grew over the five-year window.
- **Volatility:** This tracks the standard deviation of daily returns. It shows the risk and how wildly the portfolio value swings.
- **Sharpe Ratio:** This measures risk-adjusted return. It shows if the extra returns were worth the added volatility.
- **Max Drawdown:** This tracks the largest drop from a peak to a trough. It highlights the worst-case loss during a market dip.
- **Alpha ( $\alpha$ ):** This represents the true outperformance of my strategy. It captures the extra return compared to the benchmark ETF. It proves if my stock selection added unique value.

## Multi-Sector Automation ("The Smarts")

I automated this entire process in Python. Instead of writing rigid code for each individual sector, a loop scans the whole market at once.

The logic flows like this:

```
For each Sector ETF in my target list:  
    For each Rebalancing_Period in the 5-year window:  
        1. Find the historical ETF stocks for this date  
        2. Filter and rank them to select the top K stocks  
        3. Calculate the profit and log the total returns  
        4. Record the Ticker Delta (stocks added or removed)
```

The script automatically runs this loop across multiple sectors. I also added a feature to track the delta. This shows the exact list of stocks added or removed at each step. It is a great debugging tool. I can see exactly why the script bought or sold a company. This makes the strategy completely transparent.

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## Experimental Design

I set up my backtests as an experiment with adjustable settings. By changing these variables, I can see which combinations protect the portfolio and which ones add risk. I focused on testing three main categories:

- **Metric combinations:** I test the Price-to-Book (P/B) ratio by itself. Then I test it combined with profit guardrails. These require stocks to meet a minimum Return on Assets (ROA) and Return on Equity (ROE) before I rank them.
- **Basket size (K):** I try different portfolio sizes like 5, 10, 20, or 50 stocks. This helps me find the right balance between choosing only the cheapest stocks and spreading out the risk.
- **Sector percentiles:** Financial metrics vary a lot between sectors like Tech and Utilities. To fix this, I rank companies against their own sector peers. This lets me filter out the bottom 10% or 20% of unusual or tiny stocks before ranking the final basket.

**Experiment Goal:** My goal is not to find a single perfect setting that makes the most money. I want to see the broader trends. For example, I want to check if a larger basket lowers volatility, or if an ROE filter protects the portfolio during market drops.

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## Results of Pure P/B Selector

I wanted to see what happens with a pure Price-to-Book strategy. I bought the cheapest stocks in the market. I used no safety nets, no quality filters, and no size limits.

I ranked the stocks by P/B. I tested four different portfolio sizes: 5, 10, 50, and 100 stocks. I also ran tests with two control groups. One control group bought the largest companies by market cap. The other group used an equity-based strategy. I equally weighted all portfolios. Then I compared them to their sector benchmarks.

## Core Analysis

### Impact of Basket Size

Basket size had a direct impact on risk. This proves that diversification works. In every sector I checked, a bigger basket meant lower volatility.

The 5-stock baskets were highly volatile and unstable. Volatility dropped when I expanded the size to 50 and 100 stocks. A larger basket protects capital from single-company wild swings.

### Impact of Ranking

The way I ranked the stocks changed both my risk and my returns:

- **The Value Trap:** Ranking by cheap P/B alone led me straight to dying businesses. Without quality checks, the system just picks highly distressed companies.
- **Large-Cap Safety:** In small 5-stock portfolios, picking the biggest companies by market cap was much safer. Big companies protect against sudden failures. My pure P/B ranking only beat the big companies in Tech and Consumer Defensive.

### Behavioral Differences Across Sectors

The strategy performed differently across different market sectors:

- **The outliers:** In Basic Materials, the 5-stock portfolio showed huge returns but extreme volatility. A few distressed penny stocks swung wildly and distorted the average.
- **Total failure:** In Healthcare and Consumer Cyclical, the pure value strategy lost heavy capital. The concentrated Healthcare portfolio fell into value traps and crashed deeply.
- **Winners:** Financial Services and Technology fit the strategy well. The P/B metric caught real mispricings. It beat both large caps and the sector benchmarks.

### Other Interesting Findings

In small sectors like Utilities, the 50-stock and 100-stock portfolios had identical performance data.

This happens because those sectors have very few available stocks. Sorting by different metrics still results in buying the exact same pool of companies.

Across all 11 sectors, the pure P/B strategy beat the benchmark's annual return. However, it forced me to endure double the volatility.

### Performance Matrix

The table below shows the results. It compares the pure P/B strategy and the market cap control against the benchmarks.

| Sector / Strategy | Basket Size (K) | Ranking / Selection Metric | CAGR | Volatility | Sharpe Ratio | Max Drawdown |
|-------------------|-----------------|----------------------------|------|------------|--------------|--------------|
|                   |                 |                            |      |            |              |              |

|                                             |     |                  |                |                |             |         |
|---------------------------------------------|-----|------------------|----------------|----------------|-------------|---------|
| <b>Basic Materials</b><br>( <i>XLB</i> )    | —   | <b>Benchmark</b> | 9.17%          | 19.50%         | 0.42        | -35.20% |
|                                             | 5   | P/B              | <b>147.59%</b> | <b>838.59%</b> | 0.17        | -50.12% |
|                                             | 5   | mCap             | 10.53%         | 23.62%         | <b>0.36</b> | -40.21% |
|                                             | 50  | P/B              | <b>61.50%</b>  | 85.08%         | <b>0.70</b> | -20.45% |
|                                             | 50  | mCap             | 16.24%         | 22.94%         | 0.62        | -23.78% |
|                                             | 100 | P/B              | <b>42.07%</b>  | 49.98%         | <b>0.80</b> | -20.32% |
|                                             | 100 | mCap             | <b>42.07%</b>  | 49.98%         | <b>0.80</b> | -20.32% |
| <b>Consumer Cyclical</b><br>( <i>XLV</i> )  | —   | <b>Benchmark</b> | 9.47%          | 18.20%         | 0.47        | -32.10% |
|                                             | 5   | P/B              | -15.77%        | 66.85%         | -0.27       | -90.32% |
|                                             | 5   | mCap             | <b>17.15%</b>  | <b>32.69%</b>  | <b>0.46</b> | -46.46% |
|                                             | 50  | P/B              | <b>7.76%</b>   | 32.68%         | <b>0.18</b> | -43.11% |
|                                             | 50  | mCap             | 5.39%          | <b>21.95%</b>  | 0.15        | -39.49% |
|                                             | 100 | P/B              | <b>6.87%</b>   | 27.55%         | <b>0.18</b> | -38.28% |
|                                             | 100 | mCap             | 6.09%          | <b>22.23%</b>  | <b>0.18</b> | -39.21% |
| <b>Healthcare</b><br>( <i>XLV</i> )         | —   | <b>Benchmark</b> | 6.98%          | 14.50%         | 0.41        | -25.40% |
|                                             | 5   | P/B              | -19.53%        | 58.16%         | -0.37       | -83.04% |
|                                             | 5   | mCap             | <b>14.67%</b>  | <b>26.20%</b>  | <b>0.48</b> | -21.14% |
|                                             | 50  | P/B              | <b>8.07%</b>   | 40.57%         | <b>0.15</b> | -60.65% |
|                                             | 50  | mCap             | 3.87%          | <b>17.53%</b>  | 0.11        | -27.33% |
|                                             | 100 | P/B              | <b>5.17%</b>   | 33.89%         | <b>0.09</b> | -55.16% |
|                                             | 100 | mCap             | -0.38%         | <b>17.21%</b>  | -0.14       | -30.67% |
| <b>Technology</b><br>( <i>XLK</i> )         | —   | <b>Benchmark</b> | 18.69%         | 22.10%         | 0.80        | -30.50% |
|                                             | 5   | P/B              | <b>30.72%</b>  | 66.09%         | <b>0.43</b> | -61.75% |
|                                             | 5   | mCap             | 8.97%          | <b>27.45%</b>  | 0.25        | -40.99% |
|                                             | 50  | P/B              | <b>44.31%</b>  | 65.98%         | <b>0.64</b> | -27.79% |
|                                             | 50  | mCap             | 13.41%         | <b>24.26%</b>  | 0.47        | -39.38% |
|                                             | 100 | P/B              | <b>32.19%</b>  | 48.56%         | <b>0.62</b> | -30.94% |
|                                             | 100 | mCap             | 10.38%         | <b>23.29%</b>  | 0.36        | -35.57% |
| <b>Financial Services</b><br>( <i>XLF</i> ) | —   | <b>Benchmark</b> | 19.92%         | 20.50%         | 0.92        | -35.80% |
|                                             | 5   | P/B              | <b>10.46%</b>  | 27.82%         | 0.30        | -32.50% |
|                                             | 5   | mCap             | 8.41%          | <b>17.49%</b>  | <b>0.37</b> | -35.61% |
|                                             | 50  | P/B              | <b>20.05%</b>  | <b>20.48%</b>  | <b>0.88</b> | -20.23% |
|                                             | 50  | mCap             | 17.39%         | 21.05%         | 0.73        | -32.65% |
|                                             | 100 | P/B              | <b>19.11%</b>  | <b>17.26%</b>  | <b>0.99</b> | -14.97% |
|                                             | 100 | mCap             | 15.25%         | 19.12%         | 0.69        | -25.94% |

|                                   |     |                  |               |               |             |         |
|-----------------------------------|-----|------------------|---------------|---------------|-------------|---------|
| <b>AGGREGATE</b><br>(All Sectors) | —   | <b>Benchmark</b> | 12.85%        | 17.16%        | 0.69        | -31.80% |
|                                   | 5   | P/B              | <b>30.69%</b> | 213.50%       | 0.14        | -63.55% |
|                                   | 5   | mCap             | 11.95%        | <b>21.49%</b> | <b>0.42</b> | -36.88% |
|                                   | 50  | P/B              | <b>27.20%</b> | 48.96%        | <b>0.51</b> | -34.45% |
|                                   | 50  | mCap             | 12.87%        | <b>19.55%</b> | 0.49        | -31.73% |
|                                   | 100 | P/B              | <b>21.11%</b> | 34.64%        | <b>0.55</b> | -31.93% |
|                                   | 100 | mCap             | 18.67%        | <b>22.58%</b> | 0.47        | -32.11% |

## Summary & Next Steps

My first test showed that buying stocks just because they have a cheap Price-to-Book ratio is too risky. It works well in Tech and Financials, but it leaves me completely exposed to value traps and wild price swings in other sectors.

To fix these problems, I need to make a few changes in the next steps of my project:

1. First, I will set a fixed group of stocks so the portfolio doesn't run out of options (Section 8).
2. Second, I will add strict quality checks using Return on Assets (ROA) and Return on Equity (ROE). This will filter out broken companies before I even look at their prices (Section 9).

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## “Clean” P/B test

Next, I wanted to see how the Price-to-Book strategy works when I get rid of tiny companies, weird outliers, and businesses that barely own anything.

To do this, I set up a few basic rules before picking any stocks:

- **Market Cap:** I dropped the smallest 25% of companies.
- **Book Value:** I dropped the bottom 50% of companies based on Book Value.
- **P/B Cutoff:** I only looked at the “cheaper” (low P/B) half of the remaining market.
- **Price Floor:** I set a \$1 minimum stock price to avoid penny stocks.

I calculated these percentiles for each sector separately so the comparison stayed fair. Just like before, I tested different portfolio sizes and compared everything to sector benchmarks.

## Core Analysis

### Impact of Basket Size

The results from this clean run confirm a basic rule: a bigger basket means lower volatility. Even with my filters, a tiny 5-stock portfolio was still too volatile. But as I added more stocks, the volatility dropped across the board.

I saw this clearly in Financial Services. With just 5 stocks, volatility was a bumpy 50.31%. But when I expanded the basket to 100 stocks, volatility dropped to a steady 17.12%, and the Sharpe Ratio jumped to 0.99.

### Impact of Ranking

Applying these baseline filters fundamentally shifted how our ranking metrics interacted with the underlying asset pool:

- **Noticeable gains:** In healthier sectors, filtering out micro-cap distress allowed the value ranking to isolate genuinely mispriced, high-yielding opportunities. At moderate basket sizes ( $K=10$ ), the pure Price-to-Book ranking achieved massive outperformance over the size-based control group in Technology, Energy, and Basic Materials.
- **Large-Cap Safety:** Despite the cleaner sandbox, selecting the largest companies by Market Capitalization still acted as a powerful safety shield in structurally weak or cyclical sectors. In highly concentrated portfolios, chasing large caps consistently yielded better risk-adjusted returns and shallower drawdowns than blindly picking the cheapest remaining options.

### Behavioral Differences Across Sectors

How healthy a sector is determines how well this value strategy works:

- **More Realistic Returns:** The biggest surprise was in Basic Materials. In my first raw test, the 5-stock portfolio showed a crazy 147.59% return. In this clean test, that dropped to just 2.73%. This proves the huge earlier return was just an illusion caused by tiny, volatile stocks that my size filter now blocks.
- **Persistent Value Traps:** Healthcare and Consumer Cyclical were still a mess. Even after removing tiny companies, small portfolios in these sectors lost a lot of money. They posted deep losses of -25.22% and -18.40%. A cheap P/B ratio alone can't save a dying sector.
- **Winners:** Financial Services, Tech, and Energy worked beautifully. For example, a 10-stock Tech portfolio gave a fantastic 56.32% annual return with a strong 1.07 Sharpe Ratio, easily beating the benchmark and the large-cap groups.

### Other Interesting Findings

One interesting find is that the ranking metric matters less as the basket grows. In my first raw test, the different ranking strategies only gave the same results when I hit a 100-stock basket. In this clean test, sectors like Utilities and Business Services showed identical performance as early as the 50-stock mark.

**Why this happened:** Since my new filters throw out the smallest 25% companies up front, the pool of available stocks shrinks. In smaller sectors, fewer than 50 companies pass the basic checks. So no matter how I rank them, the portfolio ends up buying the exact same surviving stocks.

## Cleaned Baseline Performance Matrix

| Sector / Strategy                  | Basket Size (K) | Ranking / Selection Metric | CAGR          | Volatility    | Sharpe Ratio | Max Drawdown   |
|------------------------------------|-----------------|----------------------------|---------------|---------------|--------------|----------------|
| <b>Basic Materials</b><br>(XLB)    | —               | <b>Sector Benchmark</b>    | 9.17%         | 19.50%        | 0.42         | -35.20%        |
|                                    | 5               |                            | 2.73%         | 32.07%        | 0.02         | -42.65%        |
|                                    | 5               | P/B                        | 10.53%        | <b>23.62%</b> | <b>0.36</b>  | -40.21%        |
|                                    | 50              | mCap                       | <b>25.04%</b> | 30.38%        | <b>0.76</b>  | <b>-22.83%</b> |
|                                    | 50              | P/B                        | 16.30%        | <b>22.92%</b> | 0.62         | -23.78%        |
|                                    | 100             | mCap                       | <b>19.76%</b> | 24.82%        | <b>0.72</b>  | -22.60%        |
|                                    | 100             | P/B<br>mCap                | <b>19.76%</b> | <b>24.82%</b> | <b>0.72</b>  | -22.60%        |
| <b>Consumer Cyclical</b><br>(XLY)  | —               | <b>Sector Benchmark</b>    | 9.47%         | 18.20%        | 0.47         | -32.10%        |
|                                    | 5               |                            | -18.40%       | 64.29%        | -0.32        | -90.82%        |
|                                    | 5               | P/B                        | <b>17.15%</b> | <b>32.69%</b> | <b>0.46</b>  | <b>-46.46%</b> |
|                                    | 50              | mCap                       | 4.00%         | 31.38%        | 0.06         | -48.19%        |
|                                    | 50              | P/B                        | <b>5.39%</b>  | <b>21.95%</b> | <b>0.15</b>  | <b>-39.49%</b> |
|                                    | 100             | mCap                       | 4.55%         | 27.09%        | 0.09         | -39.75%        |
|                                    | 100             | P/B<br>mCap                | <b>6.09%</b>  | <b>22.23%</b> | <b>0.18</b>  | <b>-39.21%</b> |
| <b>Healthcare</b><br>(XLV)         | —               | <b>Sector Benchmark</b>    | 6.98%         | 14.50%        | 0.41         | -25.40%        |
|                                    | 5               |                            | -25.22%       | 58.03%        | -0.47        | -89.78%        |
|                                    | 5               | P/B                        | <b>14.92%</b> | <b>26.66%</b> | <b>0.48</b>  | <b>-21.14%</b> |
|                                    | 50              | mCap                       | -6.03%        | 36.33%        | -0.22        | -62.44%        |
|                                    | 50              | P/B                        | <b>3.90%</b>  | <b>17.56%</b> | <b>0.11</b>  | <b>-27.33%</b> |
|                                    | 100             | mCap                       | -4.40%        | 33.93%        | -0.19        | -60.84%        |
|                                    | 100             | P/B<br>mCap                | <b>-0.41%</b> | <b>17.64%</b> | <b>-0.14</b> | <b>-31.40%</b> |
| <b>Technology</b><br>(XLK)         | —               | <b>Sector Benchmark</b>    | 18.69%        | 22.10%        | 0.80         | -30.50%        |
|                                    | 5               |                            | <b>26.36%</b> | 67.47%        | <b>0.36</b>  | -63.93%        |
|                                    | 5               | P/B                        | 8.97%         | <b>27.45%</b> | 0.25         | <b>-40.99%</b> |
|                                    | 50              | mCap                       | <b>20.59%</b> | 30.81%        | <b>0.60</b>  | <b>-30.21%</b> |
|                                    | 50              | P/B                        | 13.41%        | <b>24.26%</b> | 0.47         | -39.38%        |
|                                    | 100             | mCap                       | <b>16.10%</b> | 28.99%        | <b>0.49</b>  | <b>-31.69%</b> |
|                                    | 100             | P/B<br>mCap                | 10.38%        | <b>23.29%</b> | 0.36         | -35.57%        |
| <b>Financial Services</b><br>(XLF) | —               | <b>Sector Benchmark</b>    | 19.92%        | 20.50%        | 0.92         | -35.80%        |
|                                    | 5               |                            | <b>33.05%</b> | 50.31%        | <b>0.62</b>  | <b>-23.57%</b> |
|                                    | 5               | P/B                        | 8.41%         | <b>17.49%</b> | 0.37         | -35.61%        |
|                                    | 50              | mCap                       | <b>18.87%</b> | <b>20.32%</b> | <b>0.83</b>  | <b>-20.63%</b> |
|                                    | 50              | P/B                        | 17.39%        | 21.05%        | 0.73         | -32.65%        |
|                                    | 100             | mCap                       | <b>18.91%</b> | <b>17.12%</b> | <b>0.99</b>  | <b>-15.21%</b> |

|                           |     |                  |               |               |             |                |
|---------------------------|-----|------------------|---------------|---------------|-------------|----------------|
|                           | 100 | P/B<br>mCap      | 15.62%        | 19.07%        | 0.71        | -25.94%        |
| <b>Utilities</b><br>(XLU) | —   | <b>Sector</b>    | 10.29%        | 13.90%        | 0.65        | -24.80%        |
|                           | 5   | <b>Benchmark</b> | 6.65%         | 21.44%        | 0.22        | -34.48%        |
|                           | 5   | P/B              | <b>10.93%</b> | <b>15.67%</b> | <b>0.57</b> | <b>-23.27%</b> |
|                           | 50  | mCap             | <b>10.38%</b> | <b>12.87%</b> | <b>0.65</b> | <b>-13.78%</b> |
|                           | 50  | P/B              | <b>10.38%</b> | <b>12.87%</b> | <b>0.65</b> | <b>-13.78%</b> |
|                           | 100 | mCap             | <b>10.38%</b> | <b>12.87%</b> | <b>0.65</b> | <b>-13.78%</b> |
|                           | 100 | P/B<br>mCap      | <b>10.38%</b> | <b>12.87%</b> | <b>0.65</b> | <b>-13.78%</b> |

## Next Steps

Setting up these clean percentile filters was a big step forward. It got rid of the fake, explosive returns caused by tiny stocks and lowered volatility across most sectors.

But this clean test also showed a big flaw: size and asset filters don't look at whether a company is actually making money. I am still falling into bad value traps in Healthcare and Consumer Cyclical. Next, I will test using Return on Assets (ROA) and Return on Equity (ROE) to automatically weed out unprofitable companies before my value selector touches them.

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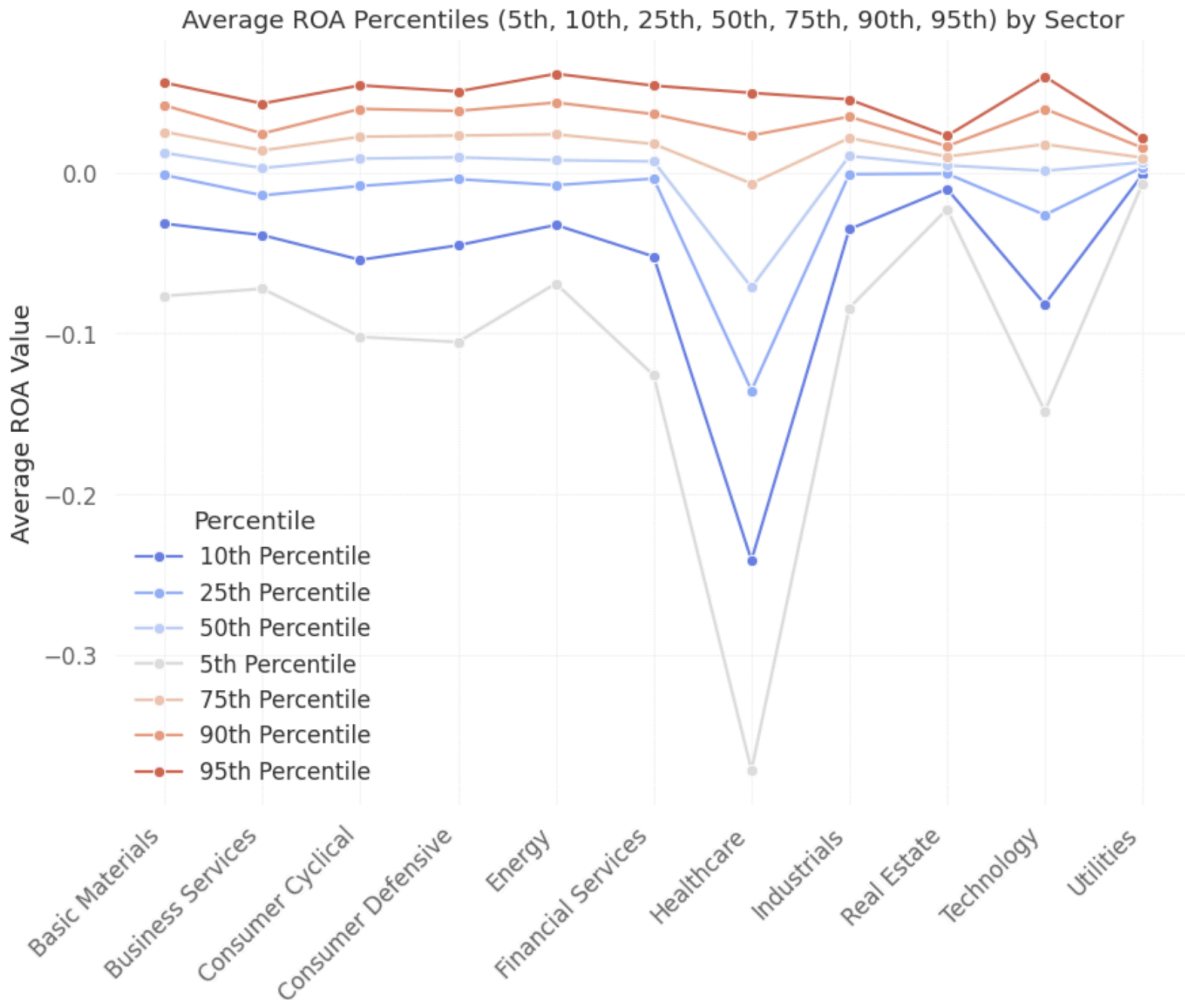
## P/B + ROA

### Objective and Setup

I added a simple rule to my basic filters. Every stock must have a positive Return on Assets of at least 0.1% before I rank it by value. ROA measures how efficiently a company uses its assets to make a profit. It divides net income by total assets:

$$\text{ROA} = \text{Net Income} / \text{Total Assets}$$

This extra check filters out unprofitable businesses. It ensures that my script only looks at companies that actually make money from their assets.



I used a strict 0.1% minimum floor for all sectors. I thought about using percentiles instead, but I decided against it. The chart above shows how percentile values look across different sectors. Allowing negative percentiles did not make sense.

In sectors where companies burn a lot of cash, like Healthcare, the median line drops well below zero. A percentile cutoff would force my script to accept unprofitable companies just to fill the list. On the other hand, stable sectors like Utilities have numbers clustered near zero. A percentile filter there would throw away healthy, profitable companies just because they are in the bottom half of a strong group. A hard absolute floor sets a clear, fair baseline for every sector.

## What Changed Relative to Earlier Runs

- Lower risk in small portfolios:** In my earlier tests, small portfolios were too volatile and risky. Adding the ROA filter changes that. Now, a small basket can deliver much higher returns without adding dangerous amounts of risk.

- **Large-cap safety:** In my past runs, buying the biggest companies was my only defense against failing businesses. With the ROA filter, that large-cap strategy became a drag on performance. It actually fell behind my pure value ranking.
- **Big turnarounds in value traps:**
  - **Consumer Cyclical (K = 5):** This sector flipped from a bad -18.40% loss and a -90.82% drawdown in the clean P/B test to an amazing **72.55% CAGR**. The maximum drawdown also shrank to just -11.32%.
  - **Technology (K = 5):** This portfolio performed exceptionally well. It reached an impressive **61.89% CAGR** with a strong 1.62 Sharpe ratio.
  - **Financial Services (K = 5):** This setting cut the maximum drawdown to a very safe **-3.87%**. At the same time, it delivered a solid **32.24% CAGR**.
- Adding strict size, asset, and earnings rules made the pool of available stocks much smaller. In tiny sectors like Utilities, the pool ran out when I set the basket size to 50. Because the script bought every surviving stock, different ranking methods yielded identical results.
- **Total outperformance:** Across all 11 sectors, the 5-stock value portfolio produced a massive **37.79% CAGR** (1.20 Sharpe). It easily beat the sector benchmark of 12.70% and the large-cap control of 11.05%.

## The Hidden Traps and Downsides

- **Healthcare is still broken at K = 5:** Even with the new profit filter, the 5-stock value portfolio lost money. It had a **-4.06% CAGR**, a negative **-0.26 Sharpe**, and a deep **-49.38% drawdown**. It fell far behind the benchmark of 6.98% and the large-cap control of 4.33%.
- **The cost of concentration:** The long-term returns are great, but the short-term swings are brutal. With just 5 stocks, volatility spiked to **56.04%** in Consumer Cyclical and **40.25%** in Basic Materials. This extreme volatility would be too risky for most real-world investors.
- **Utilities value lag:** Concentration backfired in slow-growth, regulated sectors. The 5-stock value strategy earned a **9.70% CAGR**, which fell short of the 10.29% benchmark.
- Ranking metrics stop working for large portfolios when the candidate pool is small. For example, if only 20 stocks pass my filters, setting the basket size to 50 or 100 forces the script to buy all 20 companies. Since it buys the whole pool, sorting by P/B or market cap gives the exact same portfolio. Extra returns disappear because I lose the value focus completely.

## Performance Matrix

| Sector / ETF                   | Basket Size (K) | Selection Metric | CAGR   | Volatility | Sharpe Ratio | Max Drawdown |
|--------------------------------|-----------------|------------------|--------|------------|--------------|--------------|
| <b>Basic Materials (XLB)</b>   | —               | <b>Benchmark</b> | 9.17%  | 19.50%     | 0.42         | -35.20%      |
|                                | 5               | PB               | 52.02% | 40.25%     | 1.24         | -17.95%      |
|                                | 5               | MCap             | 11.62% | 24.39%     | 0.39         | -40.21%      |
|                                | 50              | PB               | 27.12% | 21.82%     | 1.15         | -22.71%      |
|                                | 50              | MCap             | 19.11% | 22.01%     | 0.78         | -20.97%      |
|                                | 100             | PB               | 25.86% | 21.60%     | 1.10         | -20.25%      |
|                                | 100             | MCap             | 25.86% | 21.60%     | 1.10         | -20.25%      |
| <b>Consumer Cyclical (XLY)</b> | —               | <b>Benchmark</b> | 9.47%  | 18.20%     | 0.47         | -32.10%      |
|                                | 5               | PB               | 72.55% | 56.04%     | 1.26         | -11.32%      |
|                                | 5               | MCap             | 23.23% | 33.16%     | 0.64         | -44.81%      |
|                                | 50              | PB               | 21.90% | 27.25%     | 0.73         | -25.80%      |
|                                | 50              | MCap             | 11.82% | 19.55%     | 0.50         | -29.65%      |
|                                | 100             | PB               | 20.29% | 28.82%     | 0.63         | -25.77%      |
|                                | 100             | MCap             | 11.62% | 22.07%     | 0.44         | -31.52%      |
| <b>Healthcare (XLV)</b>        | —               | <b>Benchmark</b> | 6.98%  | 14.50%     | 0.41         | -25.40%      |
|                                | 5               | PB               | -4.06% | 23.75%     | -0.26        | -49.38%      |
|                                | 5               | MCap             | 4.33%  | 20.30%     | 0.11         | -22.85%      |

|                                 |     |                  |        |        |      |         |
|---------------------------------|-----|------------------|--------|--------|------|---------|
|                                 | 50  | PB               | 14.28% | 16.05% | 0.77 | -13.07% |
|                                 | 50  | MCap             | 5.75%  | 15.50% | 0.24 | -25.76% |
|                                 | 100 | PB               | 11.17% | 15.19% | 0.62 | -20.05% |
|                                 | 100 | MCap             | 10.78% | 14.33% | 0.61 | -20.64% |
| <b>Technology (XLK)</b>         | —   | <b>Benchmark</b> | 18.69% | 22.10% | 0.80 | -30.50% |
|                                 | 5   | PB               | 61.89% | 36.85% | 1.62 | -13.57% |
|                                 | 5   | MCap             | 19.48% | 26.55% | 0.66 | -38.36% |
|                                 | 50  | PB               | 30.08% | 24.62% | 1.14 | -14.94% |
|                                 | 50  | MCap             | 16.58% | 21.37% | 0.68 | -31.44% |
|                                 | 100 | PB               | 22.51% | 22.95% | 0.89 | -22.26% |
|                                 | 100 | MCap             | 15.07% | 21.08% | 0.62 | -29.76% |
| <b>Financial Services (XLF)</b> | —   | <b>Benchmark</b> | 19.92% | 20.50% | 0.92 | -35.80% |
|                                 | 5   | PB               | 32.24% | 18.82% | 1.61 | -3.87%  |
|                                 | 5   | MCap             | 6.65%  | 19.51% | 0.24 | -40.43% |
|                                 | 50  | PB               | 23.65% | 17.12% | 1.26 | -0.09%  |
|                                 | 50  | MCap             | 21.98% | 19.37% | 1.03 | -25.20% |
|                                 | 100 | PB               | 21.46% | 16.68% | 1.17 | -13.06% |
|                                 | 100 | MCap             | 20.41% | 17.25% | 1.07 | -17.33% |
| <b>Utilities (XLU)</b>          | —   | <b>Benchmark</b> | 10.29% | 13.90% | 0.65 | -24.80% |
|                                 | 5   | PB               | 9.70%  | 18.64% | 0.41 | -27.54% |
|                                 | 5   | MCap             | 7.73%  | 14.34% | 0.40 | -21.91% |

|                                |     |                        |        |        |      |         |
|--------------------------------|-----|------------------------|--------|--------|------|---------|
|                                | 50  | PB                     | 10.71% | 13.43% | 0.65 | -13.69% |
|                                | 50  | MCap                   | 10.71% | 13.43% | 0.65 | -13.69% |
|                                | 100 | PB                     | 10.71% | 13.43% | 0.65 | -13.69% |
|                                | 100 | MCap                   | 10.71% | 13.43% | 0.65 | -13.69% |
| <b>AGGREGATE (All Sectors)</b> | —   | <b>Equal-Wtd Bench</b> | 12.70% | 16.50% | 0.65 | -30.20% |
|                                | 5   | PB                     | 37.79% | 25.50% | 1.20 | -18.50% |
|                                | 5   | MCap                   | 11.05% | 22.10% | 0.40 | -31.40% |
|                                | 50  | PB                     | 22.13% | 20.50% | 1.02 | -16.00% |
|                                | 50  | MCap                   | 15.80% | 19.60% | 0.70 | -24.30% |
|                                | 100 | PB                     | 19.86% | 19.80% | 0.95 | -16.50% |
|                                | 100 | MCap                   | 18.52% | 19.20% | 0.86 | -22.10% |

## Project Progression Summary

- **Pure P/B:** Sorting by cheapness alone led straight to risky penny stocks and failing businesses.
- **Cleaned P/B:** Size and asset rules got rid of penny stocks but still ignored whether companies were losing cash.
- **Cleaned P/B + ROA (this section):** Combining value with a simple profit check weeded out dying companies. This adjustment unlocked much higher returns across the market.

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## P/B + ROE

### Core Concept: Asset Efficiency vs. Financial Leverage

Now, I look at Return on Equity (**ROE**). This metric tracks how well a company creates profits from the money shareholders invest.

$$\text{ROE} = \text{Net Income} / \text{Shareholders' Equity}$$

ROE is a very popular metric, but it has a major flaw. It is highly sensitive to a company's debt. A business can take on a lot of debt to shrink its equity base. This mathematically inflates the ROE without making any real business improvements. I added an ROE hurdle of 0.1% to see if my strategy could still find great businesses or if it would fall into these high-debt leverage traps.

Earlier, I used Return on Assets (ROA) because it looks at the whole business. In this test, I swapped ROA for Return on Equity (ROE). I wanted to see if ROE is a weaker signal because companies can game it with debt.

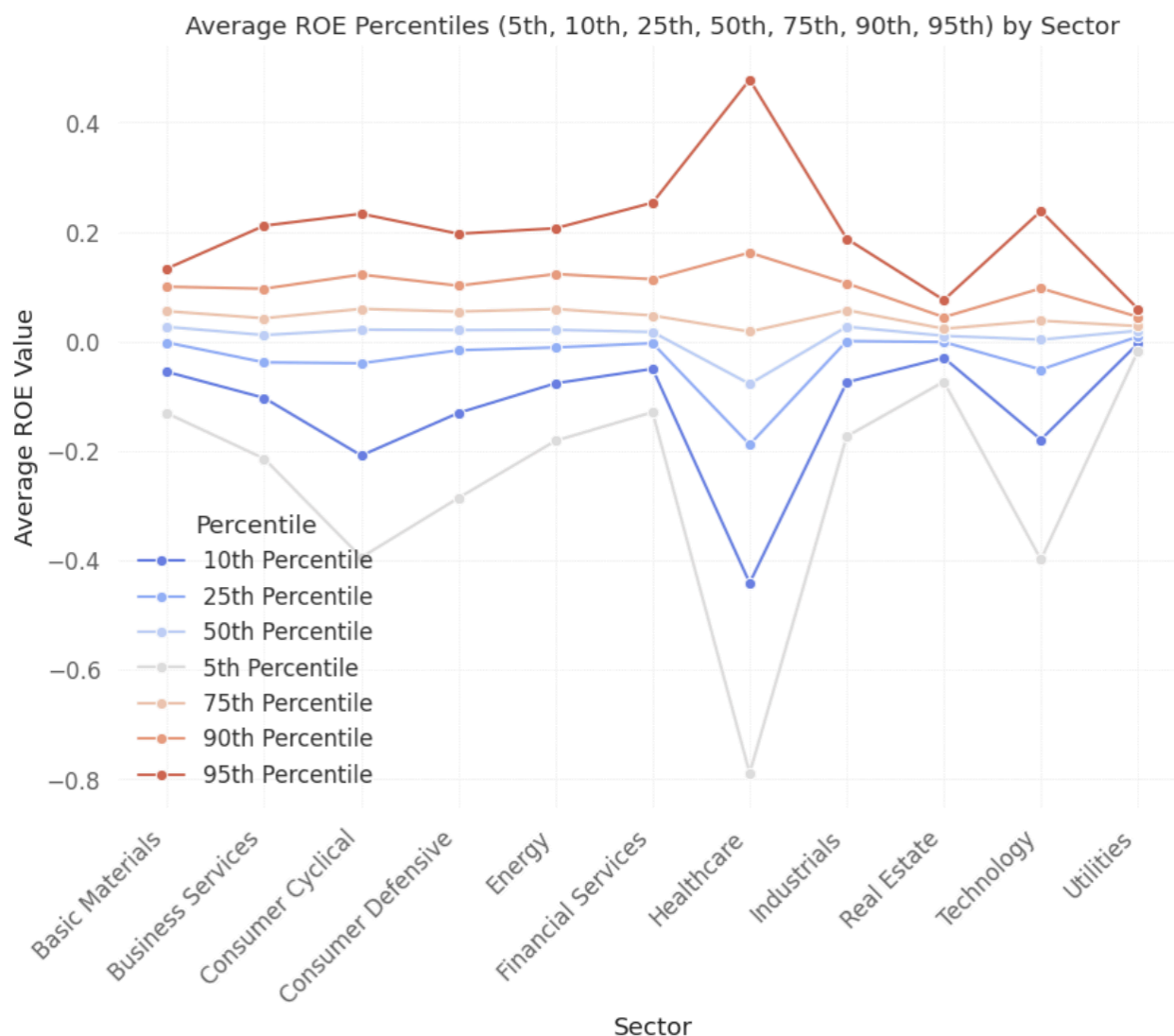
Since ROE uses equity instead of total assets, a broken business can easily hide its trouble with debt. If my strategy ignores this debt, it will buy fragile, over-leveraged companies. These are classic value traps.

### The Risk-Return Matrix: Empirical Proof of the Leverage Tax

When I looked closely at the small 5-stock portfolios, the numbers proved my suspicion. The ROE filter lowered my returns and caused much higher volatility. This tells me the strategy ended up picking risky, debt-heavy companies:

| Sector (K=5)       | ROA CAGR | ROE CAGR | CAGR Delta | ROE Volatility | ROE Sharpe |
|--------------------|----------|----------|------------|----------------|------------|
| Consumer Cyclical  | 72.55%   | 65.75%   | -6.80%     | 56.52%         | 1.13       |
| Technology         | 61.89%   | 59.18%   | -2.71%     | 36.21%         | 1.58       |
| Basic Materials    | 52.02%   | 49.64%   | -2.38%     | 41.10%         | 1.16       |
| Healthcare         | -4.06%   | -5.36%   | -1.30%     | 23.80%         | -0.31      |
| Utilities          | 9.70%    | 8.85%    | -0.85%     | 18.38%         | 0.37       |
| Financial Services | 32.24%   | 32.17%   | -0.07%     | 19.73%         | 1.53       |

I saw this clearly in Consumer Cyclical and Basic Materials. Compared to my ROA run, the ROE returns dropped significantly while volatility shot up. This is exactly what a leverage trap looks like. I took on way more risk for worse results.



## Key observations

- Financial Services was the only exception. Performance stayed almost exactly the same. Banks use leverage as a normal business tool, not a sign of trouble. Because of this, the ROE filter works perfectly there without creating weird distortions.
- Healthcare struggled in the 5-stock portfolio because many speculative biotech firms have terrible ROE numbers. They use complicated equity setups to hide their cash burn. But when I expanded the basket to 10 stocks, diversification helped water down those single-company traps. That gave me a solid 19.98% return with much smaller losses.
- In larger 50 and 100-stock portfolios, my filters ran completely out of stocks in smaller sectors like Utilities. So few companies passed the quality checks that the ranking systems completely broke down. The strategy had to buy every single surviving stock, making the portfolios identical.

## Conclusion

The ROE check does stop the system from buying bankrupt companies. However, it still gets fooled by heavy debt. For this kind of strategy, ROA is simply a better screen. It looks at real asset efficiency and ignores debt games.

# Running against broad market (SPY)

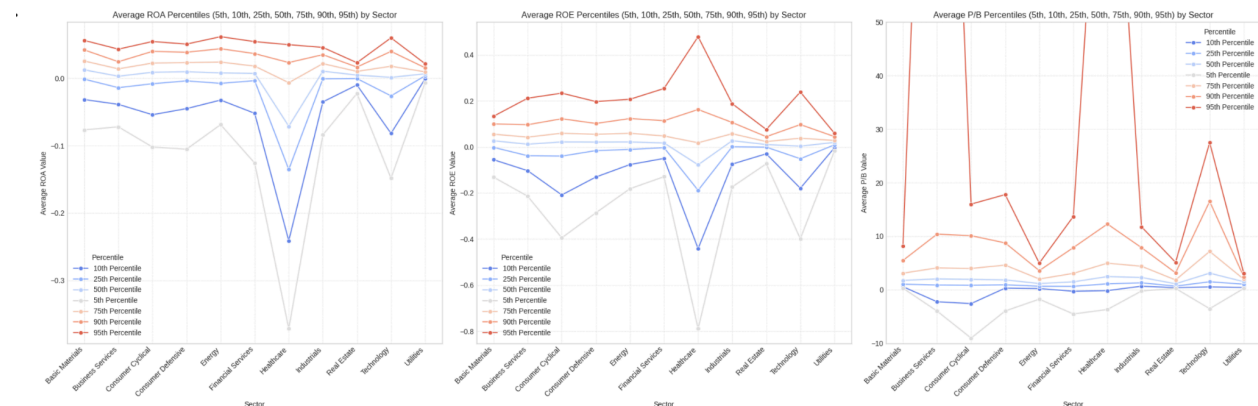
When I looked over my sector experiments, I noticed two clear patterns.

- **Small pools:** In small sectors like Utilities or Basic Materials, there were not enough companies. When I applied strict quality filters like a positive ROA, the pool dried up completely.
- **Useless ranking:** When the pool dried up, my P/B ranking stopped working. For example, if only 20 stocks passed my filters, a basket size of 50 forced the system to buy all 20 anyway. The ranking did not matter at all.

I realized I needed a larger pool of companies. This made me think about testing the entire stock market instead of separate sectors.

## Shifting to Sector-Relative Percentiles

However, I noticed that financial numbers look very different from one sector to another. I could not apply my formulas directly to the whole market.



- **Technology:** Tech companies usually have high ROA, ROE, and P/B ratios. I think this is because they do not need huge physical investments to grow.
- **Utilities:** On the other hand, Utilities are very asset-heavy. They require massive physical power grids and plants. This leads to naturally lower ROA and ROE numbers.

If I use the same cutoff for everything, my filters will break. The system will ban high-quality Tech stocks because their P/B ratios look too high. It will only buy Utilities because their ratios are naturally low.

To fix this, I created a peer-ranking strategy. I call it Sector Neutrality. Instead of comparing a Tech firm directly to a Utility firm, I rank each stock only against its direct peers. I score them on a scale from 0% to 100%. This was easy to do because my code already used sector percentiles.

#### Filters

1. **Size Protection:** First, eliminate penny stocks by requiring a minimum price of \$1.00. Then, filter out small, volatile, or illiquid companies by filtering out the bottom 25% of companies by Market Cap and the bottom 50% of companies by Book Value relative to their sector.
2. **The Quality Gates:** Keep companies that rank in the top 25% of their sector for both ROA and ROE.
3. **The Value Filter and Ranker:** I restrict my buying pool to the cheapest 25% of the sector by P/B percentile. Of the survivors that pass every single check, I rank them from cheapest to most expensive using their P/B sector percentile and buy the top K stocks.

#### Broad Market S&P 500 Backtest Results

| Strategy (Ranking Metric) | Basket Size | Return (CAGR) | Realized Volatility | Sharpe Ratio | Max Drawdown |
|---------------------------|-------------|---------------|---------------------|--------------|--------------|
| P/B Value                 | 5           | 36.84%        | 39.39%              | 0.88         | -31.19%      |
|                           | 10          | 33.61%        | 32.04%              | 0.99         | -23.73%      |
|                           | 50          | 46.02%        | 22.06%              | 2.00         | -7.34%       |
|                           | 100         | 39.02%        | 19.73%              | 1.88         | -9.67%       |
| Total Equity (Descending) | 50          | 15.45%        | 13.13%              | 1.02         | -16.62%      |
|                           | 5           | 14.21%        | 22.18%              | 0.55         | -40.09%      |
| Market Cap                | 50          | 14.61%        | 14.53%              | 0.87         | -22.26%      |
| SPY Benchmark             | —           | 15.40%        | 15.12%              | 0.89         | -23.93%      |

I usually expect a tiny 5-stock portfolio to have the highest potential returns. Spreading the risk across 50 stocks normally lowers those gains in exchange for safety.

But I noticed a strange surprise in this backtest. The 50-stock basket was not just a smoother ride. It actually beat the 5-stock basket by a wide margin. It posted a massive 46.02% annual return compared to 36.84%. Even though this is a big difference, I do not want to jump to conclusions yet. My testing window is only five years long.

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## Data Anomalies, Limitations, and Caveats

My backtest results look great on paper, but they are just simulations. In the real world, financial data is messy and trading costs money. Then there is human emotion influencing the trade decisions. These hidden risks can completely change my actual results. Here are the some problems I ran into:

### 1. Messy Data and Missing Companies

I found huge gaps in my raw dataset. My code had to throw out a massive chunk of the stock market simply because the numbers were missing.

| Metric         | Available Companies (%) | Missing Companies (%) |
|----------------|-------------------------|-----------------------|
| Sector Mapping | 6,276 (95.83%)          | 273 (4.17%)           |
| Price Data     | 5,890 (89.94%)          | 659 (10.06%)          |
| Income Data    | 3,788 (57.84%)          | 2,761 (42.16%)        |
| Balance Data   | 3,789 (57.86%)          | 2,760 (42.14%)        |

Since my strategy relies on P/B, ROA, and ROE, I had to drop any company with missing data. This created two major blind spots:

- **The Penny Stock Accident:** Over 10% of the missing companies were penny stocks. These are cheap and wildly volatile. By dropping them automatically, I accidentally cleaned up my stock pool before the strategy even started.
- **Skipping Corporate Landmines:** I hear that companies with missing data are often close to bankruptcy. By ignoring them instead of penalizing them, my code skipped major landmines. While it makes my strategy look way safer than it actually is, it also introduces the risk of “selection bias”.

### 2. The 5-Year Time Limit

I only tested this strategy for five years, from mid-2020 to mid-2025 due to simfin's freebie limits. In the stock market, five years is just a tiny snapshot.

This period was also highly unusual. It started during the COVID recovery, rode a massive tech boom, and saw interest rates spike. I did not test this through a full economic cycle like the 2008 crash. I cannot prove these returns will last. I might have just gotten lucky with a unique five-year window.

### 3. Hidden Trading Costs

I assumed that buying and selling stocks is free and instant. In real life, trading costs money and takes time.

- **Fees:** Broker commissions and taxes add up fast.
- **Price:** I assumed I could buy exactly at the adjusted closing price, which is rarely possible.
- **The 50-Stock Problem:** Managing 5 stocks is cheap. But trading 50 stocks creates a lot of extra fees that add up to sizable costs.

### 4. Variables I Didn't Test

Due to data limits, I only tested one specific setup. I left many important variables completely untouched:

- **Other Value Metrics:** I only used the P/B ratio. I do not know if P/E or EV/EBITDA would choose better stocks.
- **Other Quality Metrics:** I only checked ROA and ROE. I ignored crucial signs like total debt, and I did not test trailing averages.
- **Rebalancing Timing:** I used a rigid schedule. I do not know if rebalancing monthly or yearly would change my results.

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## Key Learnings and Takeaways

This project taught me that building an investment strategy is hard. It is not just about writing code or looking at high returns. Managing real-world data and understanding financial math and its nuances takes time and thinking.

Here are the main lessons I learned from my research. I listed them from data issues to my personal takeaways.

### 1. Data quality and cleaning

I learned that finding free and good financial data is very hard. I could not just plug the data in and expect it to work. I had to clean it first. Over 40% of my dataset was missing basic

balance sheet and income information. I had to learn how to handle these gaps. If you do not know the flaws in your data, you cannot trust your results.

## **2. Financial ratios can lie (The Leverage Trap)**

I cannot take popular metrics like Return on Equity (ROE) at face value. In my first runs, my code picked "zombie" companies. These companies had negative net income and negative equity. A negative number divided by a negative number equals a positive number. This basic math glitch made failing companies look highly profitable. I learned to check the raw numbers underneath the ratios to avoid debt traps.

## **3. Hidden anomalies can shape outcomes**

I saw how a single stock can break an entire test. In my 5-stock portfolio, a penny stock named **COGDL** jumped from \$0.01 to \$0.95 in one quarter. This huge jump messed up my portfolio returns. It made my strategy look amazing, but it was just a freak event. Now I know I must look closely at individual results to find these errors.

## **4. Diversification is important**

Finance theory says small portfolios have higher returns but more risk. But I found something different. My 50-stock portfolio did better and was safer than my 5-stock portfolio. A larger portfolio does more than just spread out market risk. It also protects against data glitches. With 50 stocks, the impact of one freak stock jump gets drowned out.

## **5. Don't compare apples to oranges**

At first, I tried using the same fixed rules for the whole market. This failed because different sectors have different baselines. A tech company has a very different balance sheet than a bank or a utility company. I learned to use **sector percentiles** instead. This allowed me to make a fair comparison.

## **6. Over-Filtering**

When I started, my filters were too strict. I used too many rigid rules and ran out of stocks to buy. This was a big problem for larger portfolios. I learned that a good strategy needs a balance. It must have quality controls, but it also needs enough stocks to build a real portfolio.

## **7. The Ultimate Tradeoff: Risk vs. Return**

This project reminded me of a classic investing rule. There is no free lunch. Spreading your risk helps smooth out market swings, but it limits your top returns. Every choice in strategy design is a trade-off. You either chase higher returns or choose peace of mind.

## **8. Brute-Forcing Parameters is Impossible (My Next Step)**

I realized I cannot test every combination of settings by hand. Changing schedules, cutoffs, and ratios takes too long because there are too many options. I recently found a technique called **Bayesian Optimization**. It acts like a smart search engine to find the best settings quickly. I plan to study this next.

## **9. Synthesizing My Interdisciplinary Interests**

My favorite part was bringing all my interests together. This project combined my accounting knowledge with my love for financial markets. It also used my skills in programming and data analysis. It was highly satisfying to see these different subjects form a working strategy.